

## OTTAWA GATINEAU, QC, Canada

WMO: 716279

Lat: 45.520N

Lon: 75.563W

Elev: 211

StdP: 14.58

Time Zone: -5.00 (NAE)

Period: 87-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.2      | -12.1 | -25.0                         | 1.4 | -15.8 | -20.7 | 1.8 | -11.0 | 23.3                  | 22.3 | 20.9 | 16.7 | 2.8                   | 0    | 0.559 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD<br>to 0.4% DB |      |
|---------------|---------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|-------------------------|------|
|               |                           | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                         |      |
|               |                           | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                    | PCWD |
| (a)           | (b)                       | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                     | (p)  |
| 7             | 23.5                      | 87.9            | 71.6 | 85.0 | 70.4 | 82.2 | 68.8 | 74.8                | 83.5 | 73.0 | 81.2 | 71.3 | 79.3 | 9.3                     | 240  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.2                            | 120.0 | 79.7 | 70.3 | 112.4 | 77.6 | 68.6 | 105.9 | 75.6 | 38.5          | 83.6 | 36.8 | 81.3 | 35.3 | 79.4 | 82.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 20.1              | 17.9 | 15.7 | -26.1                      | 92.8 | 3.0                | 1.6 | -28.2                                              | 93.9 | -30.0      | 94.9 | -31.7      | 95.8 | -33.8      | 96.9 |
|                   |      |      | -26.3                      | 78.5 | 3.0                | 2.3 | -28.4                                              | 80.2 | -30.2      | 81.5 | -31.8      | 82.8 | -34.0      | 84.4 |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 42.4   | 14.1  | 14.9  | 24.8  | 41.1  | 55.8  | 63.3  | 69.5  | 67.2  | 58.5  | 45.6  | 32.8  | 19.3  |
|                                                                                  | DBStd       | 22.01  | 13.11 | 12.17 | 11.06 | 8.63  | 8.05  | 6.72  | 5.08  | 5.72  | 8.21  | 7.97  | 9.70  | 13.70 |
|                                                                                  | HDD50       | 4876   | 1114  | 984   | 781   | 290   | 33    | 1     | 0     | 0     | 18    | 186   | 518   | 951   |
|                                                                                  | HDD65       | 8631   | 1579  | 1404  | 1246  | 719   | 304   | 110   | 16    | 44    | 227   | 602   | 965   | 1415  |
|                                                                                  | CDD50       | 2095   | 0     | 0     | 0     | 23    | 212   | 399   | 603   | 533   | 273   | 49    | 3     | 0     |
|                                                                                  | CDD65       | 373    | 0     | 0     | 0     | 1     | 17    | 58    | 154   | 112   | 31    | 0     | 0     | 0     |
|                                                                                  | CDH74       | 3962   | 0     | 0     | 0     | 28    | 284   | 728   | 1636  | 971   | 308   | 7     | 0     | 0     |
|                                                                                  | CDH80       | 1118   | 0     | 0     | 0     | 9     | 61    | 215   | 524   | 248   | 61    | 0     | 0     | 0     |
| Wind                                                                             | WSAvg       | 6.6    | 7.3   | 7.4   | 7.6   | 8.0   | 7.0   | 6.3   | 5.2   | 5.0   | 5.0   | 6.3   | 7.3   | 7.1   |
| Precipitation                                                                    | PrecAvg     | 35.7   | 2.3   | 2.3   | 2.5   | 2.9   | 3.0   | 3.5   | 3.3   | 3.2   | 3.4   | 3.1   | 3.4   | 3.0   |
|                                                                                  | PrecMax     | 43.9   | 3.8   | 5.5   | 5.1   | 5.2   | 6.3   | 7.0   | 4.9   | 8.0   | 5.5   | 5.8   | 6.1   | 4.6   |
|                                                                                  | PrecMin     | 28.4   | 0.5   | 0.1   | 0.6   | 1.2   | 1.2   | 1.2   | 1.4   | 0.4   | 1.7   | 1.1   | 1.7   | 0.9   |
|                                                                                  | PrecStd     | 4.2    | 0.9   | 1.2   | 1.3   | 1.1   | 1.4   | 1.5   | 1.0   | 1.6   | 1.3   | 1.3   | 1.1   | 0.9   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 42.9  | 44.2  | 51.9  | 77.1  | 85.9  | 90.4  | 93.0  | 89.4  | 86.2  | 73.9  | 62.0  | 47.7  |
|                                                                                  |             | MCWB   | 39.4  | 40.8  | 45.7  | 65.0  | 66.4  | 70.5  | 74.7  | 73.8  | 74.1  | 64.6  | 55.9  | 44.6  |
|                                                                                  | 2%          | DB     | 38.6  | 39.6  | 45.1  | 67.5  | 80.9  | 85.7  | 89.0  | 85.4  | 81.2  | 68.5  | 55.1  | 42.8  |
|                                                                                  |             | MCWB   | 35.9  | 35.7  | 39.5  | 53.7  | 64.9  | 70.0  | 72.1  | 71.8  | 70.4  | 60.8  | 51.1  | 40.1  |
|                                                                                  | 5%          | DB     | 35.6  | 37.1  | 41.5  | 60.6  | 76.9  | 81.9  | 85.7  | 82.8  | 77.7  | 63.5  | 50.9  | 39.1  |
|                                                                                  |             | MCWB   | 33.4  | 34.3  | 37.1  | 50.6  | 62.3  | 67.0  | 70.9  | 70.5  | 68.1  | 56.8  | 47.5  | 36.9  |
|                                                                                  | 10%         | DB     | 33.1  | 34.4  | 38.7  | 55.4  | 72.5  | 77.9  | 82.4  | 79.8  | 73.3  | 58.9  | 46.3  | 36.1  |
|                                                                                  |             | MCWB   | 31.4  | 32.6  | 35.3  | 46.8  | 60.1  | 65.0  | 68.8  | 69.0  | 65.8  | 53.3  | 43.0  | 34.1  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 40.7  | 40.9  | 48.2  | 65.4  | 71.6  | 75.5  | 78.1  | 76.4  | 75.5  | 66.7  | 56.7  | 45.6  |
|                                                                                  |             | MCDB   | 42.2  | 43.0  | 52.1  | 78.4  | 80.3  | 83.8  | 88.5  | 85.2  | 83.1  | 70.9  | 60.7  | 47.7  |
|                                                                                  | 2%          | WB     | 36.1  | 36.8  | 40.9  | 56.7  | 68.2  | 72.3  | 74.9  | 74.1  | 72.5  | 62.2  | 51.9  | 40.5  |
|                                                                                  |             | MCDB   | 37.8  | 38.9  | 43.4  | 63.1  | 77.1  | 81.7  | 84.5  | 82.2  | 78.9  | 66.2  | 54.1  | 42.6  |
|                                                                                  | 5%          | WB     | 33.8  | 34.6  | 38.1  | 51.8  | 64.5  | 69.8  | 73.0  | 72.3  | 70.0  | 58.0  | 47.3  | 37.2  |
|                                                                                  |             | MCDB   | 35.3  | 36.5  | 40.5  | 60.3  | 73.2  | 78.7  | 82.0  | 79.9  | 74.9  | 62.4  | 50.2  | 38.8  |
|                                                                                  | 10%         | WB     | 31.6  | 32.6  | 35.8  | 48.3  | 61.6  | 67.2  | 71.1  | 70.4  | 67.1  | 53.9  | 43.4  | 34.4  |
|                                                                                  |             | MCDB   | 33.2  | 34.6  | 38.3  | 53.9  | 68.9  | 74.7  | 79.8  | 77.0  | 72.5  | 57.3  | 46.4  | 35.7  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 19.7  | 20.9  | 18.2  | 19.5  | 23.3  | 22.5  | 23.5  | 21.3  | 22.4  | 18.9  | 15.1  | 15.8  |
|                                                                                  |             | MCDBR  | 17.9  | 19.7  | 18.0  | 30.2  | 30.8  | 28.8  | 27.9  | 25.1  | 26.0  | 26.0  | 21.2  | 15.6  |
|                                                                                  |             | MCWBR  | 16.8  | 17.9  | 14.2  | 18.5  | 17.1  | 14.7  | 13.7  | 13.3  | 15.8  | 18.2  | 17.8  | 14.3  |
|                                                                                  | 5% WB       | MCDBR  | 16.5  | 19.0  | 16.4  | 25.8  | 25.5  | 22.7  | 23.9  | 21.3  | 23.5  | 23.5  | 20.6  | 15.1  |
| MCWBR                                                                            |             | 16.1   | 17.7  | 14.1  | 16.9  | 15.8  | 13.3  | 13.3  | 12.2  | 15.9  | 18.1  | 18.1  | 14.4  |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.261  | 0.287 | 0.327 | 0.345 | 0.384 | 0.415 | 0.419 | 0.400 | 0.371 | 0.339 | 0.310 | 0.278 |       |
|                                                                                  | taud        | 2.365  | 2.283 | 2.322 | 2.412 | 2.347 | 2.273 | 2.269 | 2.327 | 2.398 | 2.483 | 2.489 | 2.378 |       |
|                                                                                  | Ebn at Noon | 272    | 283   | 286   | 290   | 281   | 271   | 268   | 269   | 268   | 263   | 252   | 251   |       |
|                                                                                  | Edh at Noon | 23     | 31    | 35    | 35    | 38    | 42    | 41    | 37    | 32    | 26    | 21    | 21    |       |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 443    | 756   | 1173  | 1426  | 1645  | 1791  | 1820  | 1606  | 1254  | 738   | 449   | 322   |       |
|                                                                                  | RadStd      | 54     | 71    | 102   | 141   | 160   | 137   | 150   | 76    | 103   | 65    | 52    | 24    |       |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (32 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page